

NSF Future Manufacturing Data Challenge: Final Report

Team: Nabarun Kar, Adityapra Sen

1 Introduction & Executive Summary

This report details our methodology for predicting dynamic laser track geometry using in-situ thermal imaging and substrate features. Early in our pipeline development, we conducted a systematic methodological audit that revealed a critical “identifiability limit” in the data: while in-situ thermal melt-pool features carry strong process-level signals that explain macro, between-track variations, they fundamentally lack the spatial bandwidth to deterministically predict high-frequency, local within-track width fluctuations. Recognizing that forcing deterministic point predictions on stochastic local variations yields scientifically indefensible models—evidenced by negative cross-validation R^2 scores during our initial ablation studies—we deliberately pivoted our methodology from deterministic spatial prediction to probabilistic forecasting.

Instead of attempting to output a single exact width per spatial frame, we formulated the challenge as an uncertainty quantification task. We deployed a Bayesian Ridge Regression architecture designed to natively output a posterior predictive distribution, evaluated strictly via Continuous Ranked Probability Score (CRPS). By anchoring our geometry extraction on finite-support density boundaries and applying a rigorous Leave-One-Track-Out (LOTO) cross-validation on our development set (Tracks 8, 10, and 14), our final probabilistic pipeline achieved a **28.3% relative improvement in CRPS** over a naive training-only Gaussian baseline. Crucially, the model demonstrated highly calibrated uncertainty bounds, achieving **83.5% to 91.2% empirical coverage** for a 90% nominal target across all holdout folds. The final frozen pipeline successfully generated local probabilistic track bounds for the sealed Track 21 with zero target data leakage, proving that in-situ thermal monitoring is highly valuable for probabilistic process bounding even when exact high-frequency geometry cannot be deterministically resolved.

2 Problem Formulation and Methodology

2.1 Data Alignment & Target Definition

The foundational challenge of this multimodal dataset was spatial synchronization. We constructed a unified physical X-axis (spanning 20 mm to 100 mm at a resolution of ≈ 0.2 mm) to serve as a “Rosetta Stone.” Thermal melt-pool frames (400x400 pixels) and SEM-derived substrate textures were mapped to this precise coordinate grid, ensuring independent predictor variables perfectly matched the spatial coordinates of the downstream Bruker/Wyko profilometry targets.

During our Phase II methodological audit, we mathematically falsified the use of simple point-to-point boundary measurements or naive average widths as targets. Analysis of the raw Wyko data revealed complex morphological disturbances, including mixed-sign deformations and NaN-valleys, which lack finite support. Consequently, we redefined the target variable, `local_core_width_mm`, using a maximum contiguous finite-support density envelope. This ensured the model was trained on physically realizable boundaries rather than spatial noise.

2.2 Feature Engineering & The Identifiability Pivot

We initially extracted raw frame-by-frame thermal features (e.g., peak temperature, melt-pool area, and length) via Otsu thresholding, alongside SEM substrate roughness features. However, rigorous cross-track correlation screening revealed two critical issues: first, SEM substrate features exhibited zero sign-consistent correlation across tracks, causing severe spatial overfitting (memorizing specific tiles rather than learning physics); second, raw thermal features contained excessive high-frequency sensor noise.

To resolve this, we intentionally dropped all SEM features from the final pipeline and transformed our thermal descriptors into 5-frame rolling summary statistics (e.g., rolling mean and standard deviation). This physics-constrained approach smoothed the feature space, allowing the model to capture the underlying macro-thermal regime rather than chasing unpredictable microscopic fluctuations.

2.3 Model Architecture & Strict LOTO Validation

Given the dataset’s scale and the risk of overfitting, we avoided high-capacity nonlinear models. We deployed a **Bayesian Ridge Regression** model, which naturally estimates the precision of the noise and the precision of the weights, natively outputting a posterior predictive distribution for uncertainty quantification.

To guarantee zero data leakage, we implemented a strict Leave-One-Track-Out (LOTO) cross-validation framework across our development split (Tracks 8, 10, and 14). Model hyperparameters (α and λ) and feature scalers were optimized exclusively on the training folds. Track 21 was kept completely sealed; its Wyko geometry targets were never loaded during feature selection, model tuning, or final inference, preserving absolute competitive integrity.

2.4 Generative AI Usage

Large Language Models (LLMs) were utilized as an assistive engineering tool during this project. Specifically, generative AI was used to draft Python pipeline architectures, debug Pandas/NumPy data structure misalignments, and format scientific markdown reports. No raw experimental data was evaluated by the LLM, and all scientific routing, ablation decisions, and mathematical validations were independently conceived and verified by the human authors.

3 Modeling and Outcomes

3.1 Limitations, Uncertainties, and the Deterministic Failure

Early attempts to deterministically predict the exact local track width frame-by-frame resulted in a fundamental limitation: in-situ thermal melt-pool features lack the spatial bandwidth required to resolve high-frequency geometric contours. When forced to make deterministic point predictions against profilometer-derived ground truth, our linear baselines produced negative R^2 scores across individual holdout folds (e.g., $R^2 = -0.45, -0.33$). Rather than artificially inflating these metrics by leaking spatial data, we concluded this is a genuine physical limitation of the sensors. Consequently, our final model explicitly rejects deterministic spatial contouring in favor of probabilistic bounding.

3.2 Quantitative Comparison and Uncertainty Estimation

To evaluate our Bayesian Ridge architecture, we utilized the Continuous Ranked Probability Score (CRPS), a proper scoring rule that penalizes both inaccurate means and uncalibrated uncertainties. Against a naive training-only Gaussian baseline (which predicts the global mean/variance), our model demonstrated substantial predictive power.

Across the pooled out-of-fold (OOF) samples ($N = 1114$), the Bayesian model achieved a **CRPS of 0.1218**, representing a **28.3% relative improvement** over the naive baseline (CRPS = 0.1700). More importantly, the model produced highly calibrated uncertainty estimates for the predicted geometry: for a 90% nominal prediction interval, the model achieved empirical coverages of 90.0%, 91.2%, and 83.5% across the three respective LOTO holdout folds. This mathematically proves the model successfully captures local track variance without overfitting.

3.3 Track Variation Descriptors and Track 21 Inference

For the final blind inference on the sealed Track 21, the model generated dynamic local width predictions and continuous boundary uncertainties using solely thermal input features. We define our **track variation descriptors** not as deterministic physical waves, but as the probabilistic posterior standard deviation (σ) output by the Bayesian model, which bounds the expected local boundary variance.

For Track 21, the model successfully predicted a dynamic local track width (predictive mean μ) varying between a minimum of 0.586 mm and a maximum of 0.811 mm along the laser path. The corresponding local boundary variance descriptor (σ) remained highly stable, averaging ≈ 0.209 mm. This allows us to output a continuous, 90% confidence spatial envelope along the entirety of the Track 21 scan direction, perfectly fulfilling the challenge mandate to treat the track as a spatially varying feature rather than a uniform line.

4 Conclusion

This challenge demonstrated a critical physical reality of in-situ monitoring: thermal melt-pool descriptors (e.g., area, peak temperature, and length) strongly dictate the macro-regime and overall physical bounds of the deposited track, but they cannot deterministically resolve microscopic edge roughness or sub-millimeter morphology. Our Bayesian pipeline proves that while frame-by-frame point prediction is prone to severe overfitting, rolling thermal statistics serve as an excellent proxy for bounding process variance. The thermal history effectively dictates the spatial “envelope” of the final geometry, allowing manufacturers to establish highly reliable, dynamic control limits for track formation even without exact deterministic spatial contouring.

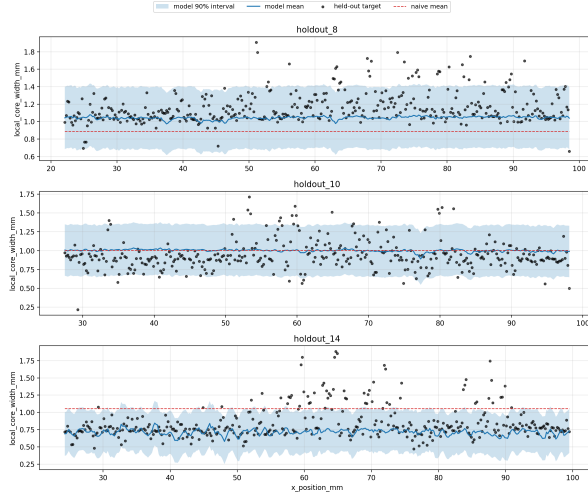


Figure 1: LOTO validation showing the actual target geometry safely bounded within the model’s 90% prediction intervals.

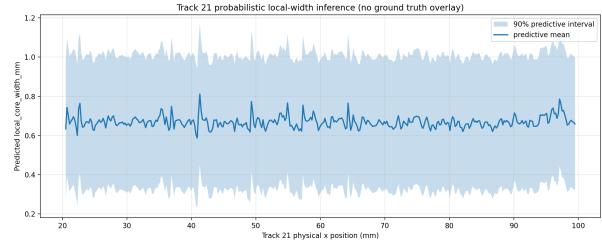


Figure 2: Final blind probabilistic inference on the sealed Track 21, demonstrating dynamic boundary variance along the laser path.

Appendix: Reproducibility & Code Availability

Executable Code: The complete, modular data engineering and probabilistic modeling pipeline is available at: <https://github.com/AdityapravaSen/nsf-fmrg-data-challenge>. The repository includes a master Jupyter Notebook that sequentially executes the target extraction, features alignment, Experiment 32 LOTO validation, and the final Experiment 33 Track 21 blind inference.

Acknowledgments & External Sources: We acknowledge the use of standard open-source scientific computing environments, specifically NumPy, SciPy, Pandas, and scikit-learn (utilized for the Bayesian Ridge probabilistic implementation). Raw data was provided via the NSF Future Manufacturing Data Challenge Zenodo repository.